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RESEARCH PAPER

Where does it come from and where does it go? Open-source flow tracing for MODFLOW 6 models

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Abstract

Groundwater studies often ask where water comes from and where it goes. Which land feeds a well, spring or stream? How much of a well's water is river water? Where does recharge under a polluted field end up? Protection zones around water supplies rest on such answers. We present open-source code, `gwsource`, that answers such questions from the flows a MODFLOW 6 model saves. The code treats the water in each model cell as fully mixed. It computes the share of each cell's water that ends at a well, a stream or another outlet. The same trace splits that outlet's water by source and computes mean travel times. The code extends the full-mixing method of FlowSource (Foley and Black 2013) to every MODFLOW 6 grid type, stream networks and linked models. In transient models it carries tagged water through storage, forward and backward in time. Every result is checked against the flow model's water budget. The outputs matched MODFLOW 6 transport of tagged water to round-off when both used the same water volumes. They also reproduced a published transport study. Particle tracking produced the same overall mean travel times, as any method that conserves water should. On a published regional model of 1.68 million cells, a trace of eight wells took 2–3 s. A trace is therefore cheap enough to repeat for every realisation of an ensemble. Full-mixing areas are wider and more diffuse than particle areas, so the threshold that defines an area is a judgement. Particle tracking remains better where a sharp edge or first arrival matters.

KEYWORDS
flow tracing, full mixing, MODFLOW 6, contributing area, sources of water, travel time, transient tracking, source protection

ARTICLE IMPACT STATEMENT

Open-source code turns saved MODFLOW 6 flows into contributing areas, water sources and travel times for any well or stream, in steady-state or transient flow.

1. INTRODUCTION

Practitioners who ask where groundwater comes from and where it goes also need to know how the answers change through the year and how sure they can be of them.

Methods that answer these questions from a groundwater flow model fall into four families. Particle tracking follows particles of water along flow paths. It is the standard tool for mapping a well's contributing area. This is the land where water enters the aquifer and later reaches the well. In MODFLOW models it runs in MODPATH (Pollock 2016) or in the particle-tracking model of MODFLOW 6 (PRT; Langevin et al. 2026). Transport with tagged water assigns the water from one source a concentration of 1 and all other water a concentration of 0. A cell's concentration is then the share of its water that came from that source (Bedekar et al. 2024). In MODFLOW 6 it runs in the Groundwater Transport (GWT) Model, with one model for each source (Langevin et al. 2022). As documented, the unit-concentration method steps forward in time only. It computes the share of each cell's water that came from a source, but not the share that will later reach a well. Adjoint and age methods solve transport equations of their own (Neupauer and Wilson 1999, Goode 1996). Full-mixing flow tracing splits each cell's mixed water among its outflows.

Particle tracking has one difficulty at weak sinks. Sink terms are the boundaries that remove water from a cell, such as a pumping well or a gaining stream. A weak sink is a cell whose sink terms take only part of the cell's outflow. A particle cannot split. It must either stop at a weak sink or pass on. If it stops, all

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its water is counted as reaching the sink. If it passes, none of it is. The rule chosen changes the contributing area (Section 4).

Under full mixing, a weak sink can take just its share of the cell's water. The rest passes on to the neighbouring cells. Two published full-mixing programs apply different rules at a weak-sink well. FlowSource (Foley and Black 2013) is used to draw source protection zones in England (Environment Agency 2019). These zones restrict activities that could pollute the water a supply well pumps. FlowSource counts all the water passing through a weak-sink well's cell as the well's water. We refer to this as net accounting. For transient models, it traces each time step as a steady-state flow field. MODALL (Potter et al. 2008) applies full mixing to steady-state MODFLOW-88 models. At a weak-sink well it counts only the water the well pumps. We refer to this as gross accounting. All these methods are compared in Supporting Information, Table S1.

This paper presents open-source code, `gwsource`, that extends the full-mixing method of FlowSource (Foley and Black 2013) to MODFLOW 6. It reads every MODFLOW 6 grid type, stream networks and linked models. It uses gross accounting by default. It checks every result against the water budget. In transient models, it carries tagged water through storage, forward and backward in time. A trace of a published regional model takes seconds (Section 4).

Section 2 explains the method and its outputs. Section 3 answers three common questions in worked examples. It then compares the outputs with particle tracking, transport with tagged water and a published study. Table 1, at the end of Section 3, summarises the ten example notebooks that ship with the code. Section 4 applies the code to a published regional model. Section 5 sets out the trade-offs of full mixing and draws the conclusions.

2. METHOD

Overview. This section explains how the code traces water and what each of its outputs means. A backward trace follows the water of a destination, such as a well, back to where it entered the model. A forward trace follows the water of a source, such as recharge, to where it leaves. Both work on the cell-by-cell flows that a MODFLOW 6 model saves, for one time step or for every step of a transient model. The backward fraction F maps where a sink's water comes from. A contributing area follows from it (example A in Section 3). The forward fraction G maps where a source's water goes. A trace also splits a sink's water by source, for example into recharge and river water. With a porosity, a trace also computes mean travel times (example C). For transient models, the code either traces each time step on its own flows or carries tagged water through storage (example B). The derivations and the solution of the equations are set out in the Supporting Information (Supporting Information, Text S1).

Full mixing and the cell budget. Every output of the code rests on the assumption of full mixing. Full mixing treats each active model cell as one well-mixed volume of water. All the water leaving a cell then has the same mix of origins and the same mix of fates, whichever outflow it takes. At the edge of a contributing area, the water of a cell then splits between the sink and other outlets, where particles from the same cell would mostly either reach the sink or miss it. For this reason, areas traced under full mixing are wider and more diffuse than particle areas. Larger cells widen them further (Section 5).

The code reads the cell-to-cell flows (FLOW-JA-FACE; Langevin et al. 2017) and the source and sink terms from the MODFLOW 6 budget file. The flow from cell i to its neighbour j is $q_{ij} \geq 0$. Source terms, such as recharge or leakage from a river, bring water into a cell. Sink terms, such as wells or drains, take it out. Each term is kept separate by package and by boundary, so that a trace can follow the water of individual boundary conditions. A stream reach, a lake or a multi-aquifer well can stay a source or sink term of the aquifer cells it exchanges water with. It can also be traced as a cell of its own, from its package budget file, so that its flows to the aquifer, to other reaches and through the water mover (MVR) become flows q_{ij} like any other. The cells of models linked by exchanges are treated as one set of cells. The choice between a term and a cell changes how a trace splits a sink's water by source (example C in Section 3).

If Src_i and Snk_i are the sums of the source and sink terms of cell i, its total inflow, total outflow and cell budget error are

$$I_i = \sum_j q_{ji} + \mathrm{Src}_i, \qquad O_i = \sum_j q_{ij} + \mathrm{Snk}_i, \qquad r_i = I_i - O_i. \quad (1)$$

The cell budgets of a MODFLOW model do not balance exactly, because the solver stops once the heads and residuals meet convergence tolerances. The cell budget error r_i is usually small but not zero. The code reports this error but never forces it to zero. Because F is written with each cell's outflow and G with its inflow, the error enters the water-budget checks that link them (below).

The backward fraction F. The backward fraction answers the question of where the water of a sink comes from. We refer to the sink terms that a trace follows as the destination. The destination can be one well, one stream reach or a group of wells traced together. The backward fraction F_i is the share of the water leaving cell i that later leaves the model through the destination. Let d_i be the outflow of cell i through the destination, and $P_{ij} = q_{ij}/O_i$ the share of its outflow that goes to cell j. Under full mixing, the water that cell i sends to cell j shares the fate of all the water leaving j. In volumes and in shares, this reads

$$F_i\, O_i = \sum_j q_{ij}\, F_j + d_i, \qquad F_i = \sum_j P_{ij}\, F_j + \frac{d_i}{O_i}. \tag{2}$$

A cell's F is the share of its outflow that leaves through the destination directly, plus the outflow-weighted sum of the F of its downstream neighbours. Eq. (2) holds in every cell, so one solve of these linear equations yields the fractions to every destination group at once. Where water circulates among cells, as it can with the XT3D option of MODFLOW 6 for anisotropic flow (Provost et al. 2017), the code solves each such flow loop exactly. FlowSource instead deletes the smallest flow in each loop (Foley and Black 2013; Supporting Information, Text S1).

A map of F shows where a destination's water comes from. A contributing area is the set of cells where F exceeds a threshold that the user chooses, such as 0.5. Because F takes in-between values at the edge of an area, the choice of threshold is a judgement (Section 5). The sum of F times the cell area yields a size of the area without a threshold. We refer to it as the expected area.

Fig. 1a works through an eight-cell example in which water enters at the west edge and as recharge and leaves through a well and a river. The well cell receives 11 m^3/d, of which its well pumps 6 m^3/d. The other 5 m^3/d flow on to the river without returning to the well. Eq. (2) then yields F = 6/11 = 0.55 in the well cell. All the water of the two cells west of the well cell flows into it, so they share its F. The cells of the upper row send part of their water past the well to the river, so their F is lower.

Gross and net accounting at weak sinks. The well cell of Fig. 1a is a weak sink, because its well takes only part of the cell's outflow (Section 1). At a weak sink the method needs a rule for how much of the cell's water to attribute to the sink. Eq. (2) applies gross accounting (Section 1). Each sink term takes its share of the cell's total outflow. Only that share counts as the sink's water, while the rest of the cell's outflow passes on to its neighbours. Gross accounting is the default because it attributes to each sink only the water that the sink takes from the model, as the MODFLOW 6 Groundwater Transport (GWT) Model does (Langevin et al. 2022), so that the water traced to a well adds up to what it pumps.

Under net accounting, the destination cells are fixed at F = 1. All the water that flows into them from other cells then counts as the destination's water. This is how FlowSource attributes water at a well (Foley and Black 2013; Supporting Information, Text S1). In Fig. 1a, net accounting adds the 5 m^3/d that flow on through the well cell to the river. It leaves out the 1 m^3/d of recharge on the well cell itself, because it works from the cell-to-cell flows only. The well is therefore attributed the 10 m^3/d that flow into its cell from its neighbours, against the 6 m^3/d it pumps. The difference is largest at a small well in a cell that carries much more water than the well pumps (Table 1, row 02).

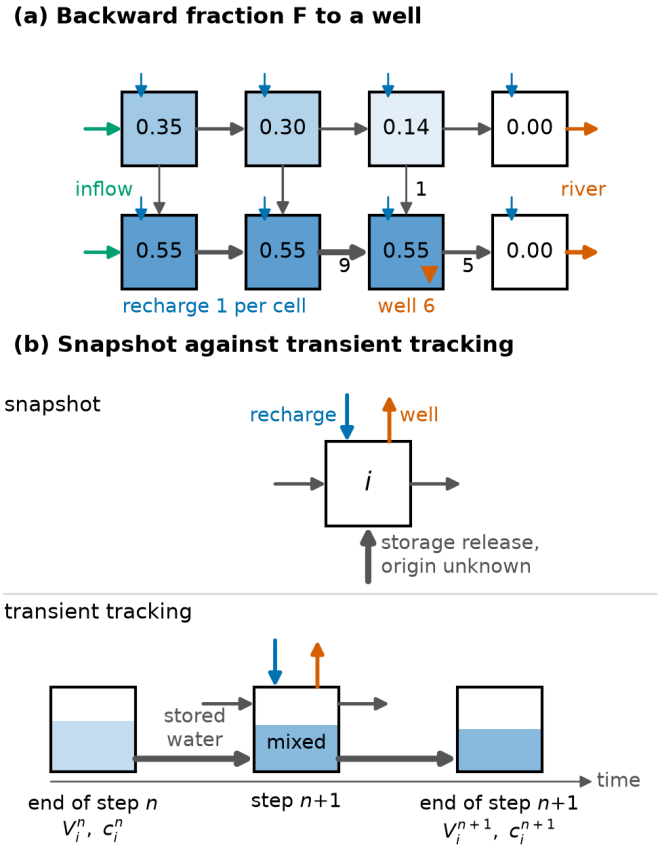


FIGURE 1 (a) Backward fractions F (in cells) to a 6 m^3/d well in an eight-cell example (gross accounting). Water enters from the west and as recharge of 1 m^3/d per cell. It leaves through the well and a river. Arrow widths show flows. Only the well cell's flows are numbered (m^3/d). In the well cell, F = 6/11 = 0.55. (b) One cell i of a transient model. A snapshot (top) traces each time step on its own flows. Water released from storage then enters the cell as a source of unknown origin. Transient tracking (bottom) carries the water stored in the cell at the end of step n into step n + 1. There it mixes with the inflows of that step. V is the cell's water volume and c the share of its water that is tagged. The height of the fill shows V and its shade shows c.

The forward fraction G. The forward fraction answers the opposite question, which is where the water of a source goes. The forward fraction G_i is the share of the water entering cell i that came from the source. Let s_i be the inflow of cell i through the source. The volume of water entering cell i that came from the source is then

$$G_i\, I_i = \sum_j q_{ji}\, G_j + s_i. \tag{3}$$

A cell's G is the share of its inflow that enters directly from the source, plus the inflow-weighted sum of the G of its upstream neighbours. Eq. (3) mirrors Eq. (2), with inflows in place of outflows. A map of G shows how much of each cell's water came from the source (Fig. 2c). G equals the concentration that GWT reaches at equilibrium on the same grid, with upstream weighting and no dispersion, when the source water carries a concentration of 1 and all other water a concentration of 0 (Table 1, row 03).

Volumes and the water-budget checks. A fraction says what share of a cell's water reaches the destination, but not how much water that is. A fraction times a flow is a volume of water per unit time. We refer to these rates as volumes. The volume through cell i is $F_i O_i$. It says how much of the water passing through the cell later reaches the destination. The volume from cell i is F_i times the cell's inflow through its source terms. It says how much water enters the model in that cell and later reaches the destination. The volumes from all cells, added up package by package, split the destination's water by source (Fig. 2b). In the same way, the volumes of a forward trace split a source's water by the sink through which it leaves.

Because they are volumes of water, the results of a backward or forward trace can be checked against the flows that MODFLOW 6 simulated. The closure check follows from adding Eq. (2) over all cells. It says that the source water traced to the destination, once corrected for the cell budget errors, adds up to the water the destination takes. The forward–backward check requires the volume from a source to a destination to be the same whether it is traced backward from the destination or forward from the source. The boundary-control check traces every sink of the model at once. It requires the fractions to all sinks to add up to 1 in every cell with outflow, so that the water of every cell leaves the model somewhere. All three checks are met to round-off for any flow field, except in loops where water circulates without ever reaching a sink (Supporting Information, Text S1). As a result, a check error larger than round-off points to a fault in the tracing. The checks test the solution of the equations. They cannot show that the flow model is right, or that full mixing describes how water moves in the aquifer.

Mean travel time. Fractions say how much water reaches a destination, but not when it arrives. Travel times, unlike fractions, need a porosity. The water volume V_i of cell i is porosity times the saturated volume of the cell. Water stays in a fully mixed cell for a mean residence time $\tau_i = V_i/O_i$, whichever outflow it then takes. Let t_i be the mean travel time from entering cell i to leaving the model through the destination, for the water that does leave there. The code solves for $m_i = F_i\, t_i$, because m_i obeys an equation of the same form as Eq. (2),

$$m_i = \tau_i F_i + \sum_j P_{ij}\, m_j. \tag{4}$$

Each cell adds its own residence time, weighted by its F, to the times of its downstream neighbours. The code reports the mean travel time $t_i = m_i/F_i$ only where F is above a small threshold (Fig. 2d), because where F is close to zero the mean rests on very little water. Mean travel times are defined for gross accounting only. Eq. (4) weights each cell's residence time by the share of its outflow that leaves through the destination. Under net accounting a destination cell has F = 1, although part of its water flows on to other sinks, so this weight no longer matches the water that leaves through the destination. A mean travel time averages fast and slow water that have mixed in the cells, so it is not a first-arrival time.

Transient flow: snapshots and transient tracking. FlowSource (Foley and Black 2013) traces a transient model one time step at a time, as if the flows of each step were steady-state (Environment Agency 2019). We refer to such a trace as a snapshot. Water released from storage then counts as a source term. Water taken into storage counts as a sink term (Fig. 1b, top). This is a limitation, because in a transient model the water that reaches a sink in one step may have entered the aquifer in an earlier one. A snapshot cannot show recharge of the wet season reaching a well in the dry season. It also cannot say where the water released from storage came from.

We extend full mixing through time. We refer to this as transient tracking. Water from the chosen sources is tagged, as a tracer would be, and carried from step to step through the water volume of each cell (Fig. 1b, bottom). The hydraulic mixing-cell method of Partington et al. (2011) carries mixed water through storage in a similar way in integrated surface water–groundwater models. Storage terms are not traced as source or sink terms. Instead they change the water volume of a cell over step n from V_i^{old} to V_i^{new}, with $V_i^{\text{old}} - V_i^{\text{new}} = \Delta t^n S_i$, where Δt^n is the length of the step and S_i the net release from storage. Let c_i^n be the share of the water in cell i at the end of step n that is tagged. With all flows taken from step n,

$$V_i^{\text{new}} c_i^n - V_i^{\text{old}} c_i^{n-1} = \Delta t^n \Big[\sum_j q_{ji}\, c_j^n + s_i^n - (O_i + r_i)\, c_i^n \Big]. \tag{5}$$

Here O_i and s_i leave out the storage terms, but r_i is still the cell budget error of Eq. (1), with the storage terms included. In Fig. 1b (bottom), V_i^n stands for V_i^{new} of step n.

Eq. (5) is the balance of tagged water in one cell over one step. The outflows carry the cell's mix at the end of the step, because the scheme is implicit in time, as in GWT (Langevin et al. 2022). The term $r_i c_i$ lets the cell budget error leave the cell at the cell's own mix, as the flow-imbalance correction of GWT does. The code applies this correction by default. Water moves by advection only, with no dispersion, diffusion, sorption or decay. Some spreading still comes from mixing whole cells and from long time steps (Supporting Information, Text S1).

Transient tracking needs a porosity and the budget of every time step. The Output Control setting SAVE BUDGET ALL writes this budget. We refer to the water in the model at the start of the simulation as initial water. It can be tagged as its own group. Two choices also affect the result. By default, the water volume of each cell starts at porosity times saturated volume and then changes only by the storage flows, so that every cell's water balance closes exactly. As an option, the volume is reset to porosity times saturated volume at the end of each step, as in GWT, so that results can be compared with a GWT run. Long implicit time steps make tagged water arrive too early. For this reason, each flow step can also be split into shorter transport sub-steps without a change to the flow model. The SI describes both options (Supporting Information, Text S1). The comparisons in Section 3 show their effect.

Backward tracking in time. Forward tracking says how much of each cell's water came from a source. It cannot say which of the water now in the aquifer will reach a sink, or by when. Backward tracking in time answers this question. It solves a balance of the same form as Eq. (5), with the roles of inflow and outflow swapped, from the last step of the simulation back to the first. This differs from MODPATH backward tracking, in which particles start at the sink and move against the flow. Backward tracking in time is the discrete counterpart of the adjoint methods that solve a transport equation backward in time (Neupauer and Wilson 1999, Cornaton and Perrochet 2006).

The backward fraction in time w_i^n is the share of the water in cell i at the start of step n, or entering it during the step, that leaves through the destination before the simulation ends. Over step n,

$$\left[V_i^{\text{new}} + \Delta t^n\,(O_i + r_i)\right] w_i^n = \Delta t^n \left[\sum_j q_{ij}\, w_j^n + d_i^n\right] + V_i^{\text{old},\,n+1}\, w_i^{n+1}. \tag{6}$$

The water in a cell during a step can reach the destination by three routes. It can leave through the destination directly, flow to a neighbour, or stay in the cell into the next step, with the volume at the start of that step. No water can reach the destination after the last step, so the solve starts there with w = 0.

A map of w at the start of the simulation shows the share of each cell's water that the destination takes before the simulation ends, for example within 30 years (Fig. 2f). A threshold on this map delineates a time-limited contributing area. The same backward run also computes the share of each cell's water that leaves through other sinks and the share still stored in the model at the end. The three shares add up to 1, apart from a remainder that the code reports (Supporting Information, Text S1). As in steady-state flow, a forward–backward check links the forward and backward runs (Supporting Information, Text S1). When the flows are held constant for long enough, c tends to the steady-state forward fraction G. In the same limit, w at the first step tends to the steady-state backward fraction F.

3. WORKED EXAMPLES

This section shows how a practitioner can use the code to answer three common questions and how far each answer can be trusted. The three worked examples ask where a well's water comes from (example A), how much of it is already in the aquifer (example B), and how much comes from each source and how old it is (example C). A paragraph before Table 1 then compares the outputs with other common approaches, such as particle tracking, transport with tagged water and a published study. The example sink is a well, but each output applies in the same way to a stream reach, a drain, a lake or a group of them. Fig. 2 shows one output of each type. Table 1, at the end of this section, summarises the eleven notebooks that ship with the code.

All examples start from one small example model. It is a single unconfined layer of 40 × 40 cells of 50 m, with recharge of 5×10^{-4} m/d on every cell. The layer is 50 m thick and has a hydraulic conductivity of 5 m/d. A constant head along the west edge brings in regional groundwater. A river runs north–south 100 m east of a supply well that pumps 800 m^3/d. Each example ships with the code as a Jupyter notebook that builds its model, reproduces its numbers and draws its figures (Section 6).

Example A: where does the well's water come from? A protection zone around a supply well starts from the land whose water reaches the well. For this reason, a practitioner needs to know how large that area is and how much to trust its edge. Fig. 2a maps F for the supply well. The size of the contributing area depends on the threshold that the user chooses. The area where F > 0.5 is smaller than the area where F > 0.1, because the higher threshold keeps only the cells that send most of their water to the well.

The edge of the area is where full mixing and particle tracking are expected to differ. The comparison with particles before Table 1 tests this edge. For the practitioner, it shows that on this model the F > 0.5 area is a fair estimate of the area that particles delineate. The F > 0.1 area serves here as a cautious outer limit. Section 4 shows that a regional model with a more complex flow system need not behave in the same way.

Example B: how much of the pumped water is already in the aquifer? A decision maker may ask how much of the water pumped over the coming decades is already in the aquifer today. The answer matters where new recharge differs in quality from the water already stored, for example where land use has changed. A seasonal version of the example model shows why snapshots cannot answer this question (Section 2). In this

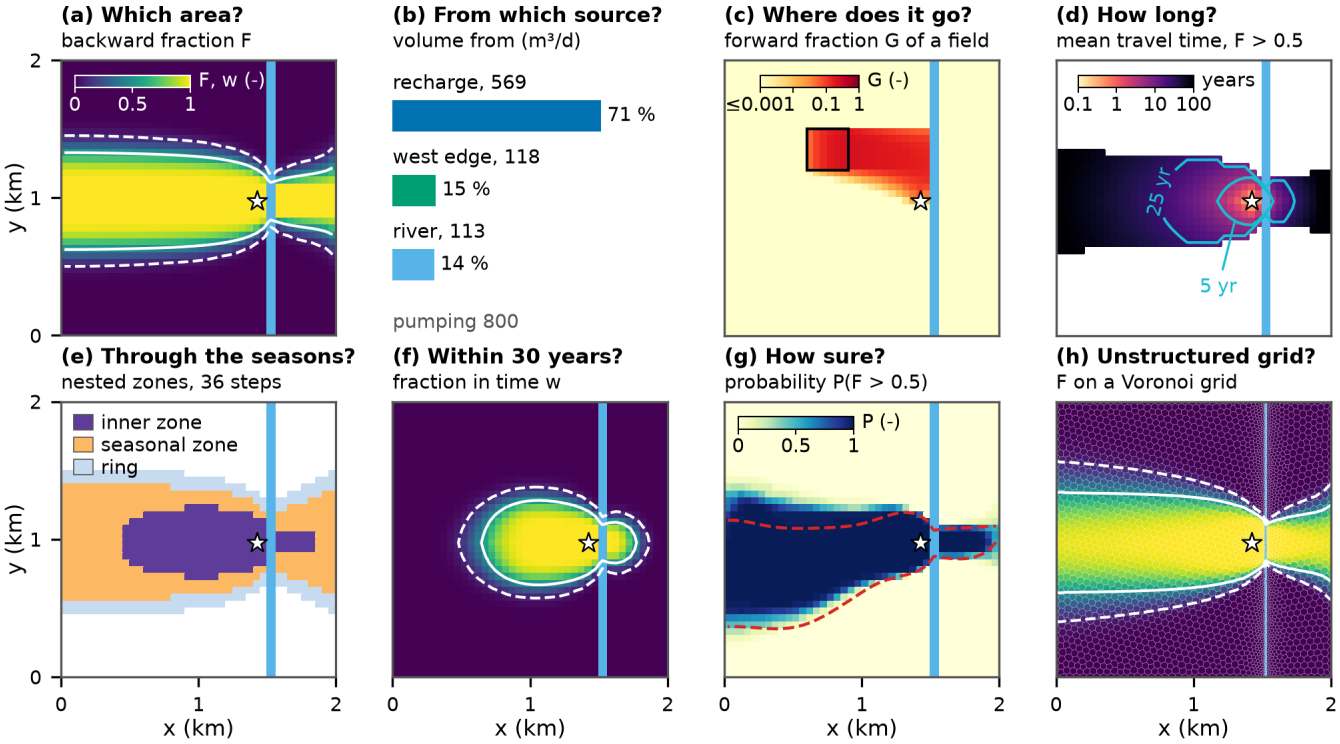


FIGURE 2 Outputs for the example well (Section 3). (a) Backward fraction F. White lines: F = 0.1 (dashed) and 0.5 (solid). (b) The well's water by source (m^3/d). (c) Forward fraction G of recharge on the outlined field. (d) Mean travel time where F > 0.5. (e) Cells with F > 0.5 in all (inner zone) or some (seasonal zone) of 36 seasonal snapshots. Ring: F > 0.1 in some snapshots. (f) Share of each cell's water pumped within 30 years. (g) Share of 50 posterior realisations with F > 0.5. Red dashed line: true F = 0.5. (h) F on a Voronoi grid. Panels (f) and (h) use the colours and lines of (a). Star: well. Blue cells: river.

version, recharge and pumping vary from season to season over 30 years, with one time step per season. Seasonal snapshots show which cells lie in the contributing area in every season and which in some seasons only (Fig. 2e; Table 1, row 05). Averaged over the 30 years, the snapshots attribute about 60 % of the well's water to recharge and 15 % to storage release (Supporting Information, Figure S1a). A snapshot cannot say where that stored water came from. It may be old water or recharge of the last wet season.

Transient tracking shows that most of the water pumped in the first decades was already in the aquifer. The tracking uses a porosity of 0.3 and four transport sub-steps per flow step. The run starts from a steady state with the well already pumping. Of all the water pumped in the first 10 years, 81 % was initial water. Over 30 years this share is 68 %. Even in the last season of year 30, 51.6 % of the well's water is initial water. In that season recharge supplies about a third of the well's water, against about 60 % averaged over the snapshots.

Backward tracking in time shows which of the water now in the aquifer reaches the well. Only the water near the well and along the main flow path reaches it within 30 years (Fig. 2f). Most of the water in the steady-state contributing area is still on its way at the end. Backward tracking attributes the same share of the water pumped over 30 years to initial water as the forward run, as the forward–backward check requires.

For the practitioner, this means that snapshots and transient tracking answer different questions. A snapshot treats the flows of each step as steady, so it says which sources feed the well but not when their water entered the aquifer. Transient tracking shows that most of the water pumped over the next decade is already in the aquifer. Where new recharge differs in quality from the stored water, we would therefore expect the well's water to change over decades rather than seasons on a model like this one.

Example C: how much water comes from each source, and how old is it? Many studies need a sink's water split by source more than the edge of its area. A water-quality study, for example, may need to know how much of a well's water is river water and how old the water from each source is. The backward trace of example A answers the first part of this question. It splits the water that the well pumps by the source term through which that water entered the aquifer. For the supply well, 71 % of the pumped water is recharge. The rest comes in about equal parts from inflow at the west edge and from river leakage (Fig.

2b; Table 1, row 01). These shares describe the mixture that reaches the well. They do not say where the rest of each source's water goes. A forward trace answers this reverse question, for example for the recharge under a field where fertiliser is spread (Fig. 2c; Table 1, row 03).

The split by source also depends on how a stream is traced. In a second version of the example model, the river becomes a stream simulated with the Streamflow Routing (SFR) package, flowing from north to south. When the stream reaches are traced as cells of their own, part of the water that would otherwise count as stream leakage turns out to be recharge that had discharged to drains or to the stream (Table 1, row 07). A study that reports how much of a well's water is stream water should therefore say whether it traced the stream reaches as cells of their own.

The backward trace also estimates the age of the well's water. With a porosity, it computes the mean travel time of each source's water to the well (Fig. 2d). This is the average time that the water takes from where it enters the aquifer to the well. On the example model, water from nearby sources such as the river arrives within a few years, while inflow at the distant west edge takes about a century (Table 1, row 06). A mean travel time is not the time at which the first water arrives. It averages fast and slow water, so it cannot say when the fastest water reaches the well. The comparison with particles before Table 1 shows which of these times depend on how a method treats mixing. For the practitioner, this means that the overall mean travel time and the split by source can be trusted more than the travel time from one cell. We would still check the split with particles where it matters (Section 4). We would also draw a short time-of-travel zone with particles (Section 5).

How the outputs compare with other methods. Practitioners may ask how the outputs compare with particle tracking and with transport of tagged water. Table 1 lists the numbers. On the flows of example A, MODFLOW 6 PRT particles that lose weight at each sink by the full-mixing shares deliver most of their water to the well from an area close in size to the F > 0.5 area. We refer to them as weight-losing particles (Supporting Information, Text S2). The F > 0.1 area holds every cell of this particle area. The two methods differ most at the edge, where most particles either reach the well or miss it while full mixing assigns the cell an in-between value (Fig. 3a). Because these particles share the full-mixing rule at sinks, they are not an independent particle reference. For example C, the same particles yield nearly the same overall mean travel time as full mixing (Table 1, row 06). This agreement is not an independent test. A particle run conserves water when the weight its particles deliver to each sink equals the water that sink takes. Any method that conserves water yields the same overall mean, because that mean equals the volume of water bound for the sink divided by the sink's inflow. The agreement therefore tests the volume attributed to the sink. The time compared with PRT is the time to entry of the sink cell, because particle trackers stop particles there (Supporting Information, Text S1). Travel times from single cells differ more, most near the edge of the area. A particle follows one path, while full mixing averages the fast and slow paths that a cell's water takes. In example B, transient tracking with GWT's water volumes matched a GWT run with tagged recharge to round-off (9.8×10^{-14}) in every cell and step. This is a consistency test, because both codes solve the same equations. The choice of water volumes changed the fractions by less than 0.004. Four transport sub-steps per flow step removed most of the early arrival of new water that long time steps cause (Supporting Information, Text S2). For an external benchmark, Bedekar et al. (2024) used transport of tagged water in MODFLOW-2005 and MT3D-USGS to show how a new well's water changes from stored water to water from its sources. Our rebuild of their model in MODFLOW 6 reproduces the published curves (Fig. 3b; Table 1, row 10). At year 25, the well's water is 48.2 % lateral inflow, 29.2 % recharge and 22.5 % stream water, against 48 %, 29 % and 23 % in the published study. The largest difference from the published curves is 1.3 percentage points, about the accuracy of values read off a published figure.

4. A REGIONAL MODEL: NEVERSINK

This section demonstrates the approach on a real-world published model. We compare the results with the published contributing areas from particle tracking and discuss the differences. The USGS Neversink/Rondout model (Fienen et al. 2022, Corson-Dosch et al. 2022) is a steady-state MODFLOW 6 model of 1.68 million cells. The published study used particle tracking to map the contributing areas of eight priority supply wells on each realisation of an ensemble of parameter sets from history matching. This section compares our results with the published area of one realisation. We refer to it as the calibrated model. We also traced every realisation of the ensemble, but those results are not presented here (Supporting Information, Text S3).

The published study mapped the wells' contributing area in the calibrated model with MODPATH 7 (Pollock 2016). One particle started at the centre of the uppermost active cell of each column within three rectangular blocks of model rows and columns, one around each cluster of priority wells (Fienen and Corson-Dosch 2021). We refer to these cells as the start cells. The 980 start cells whose particle ends in a priority well form the published contributing area. We traced the wells in the calibrated model with gross accounting. The SFR stream reaches stayed source and sink terms of the aquifer cells.

A `gwsource` trace of all the wells took a few seconds, against about 30 s for the fastest PRT run on the same flows.

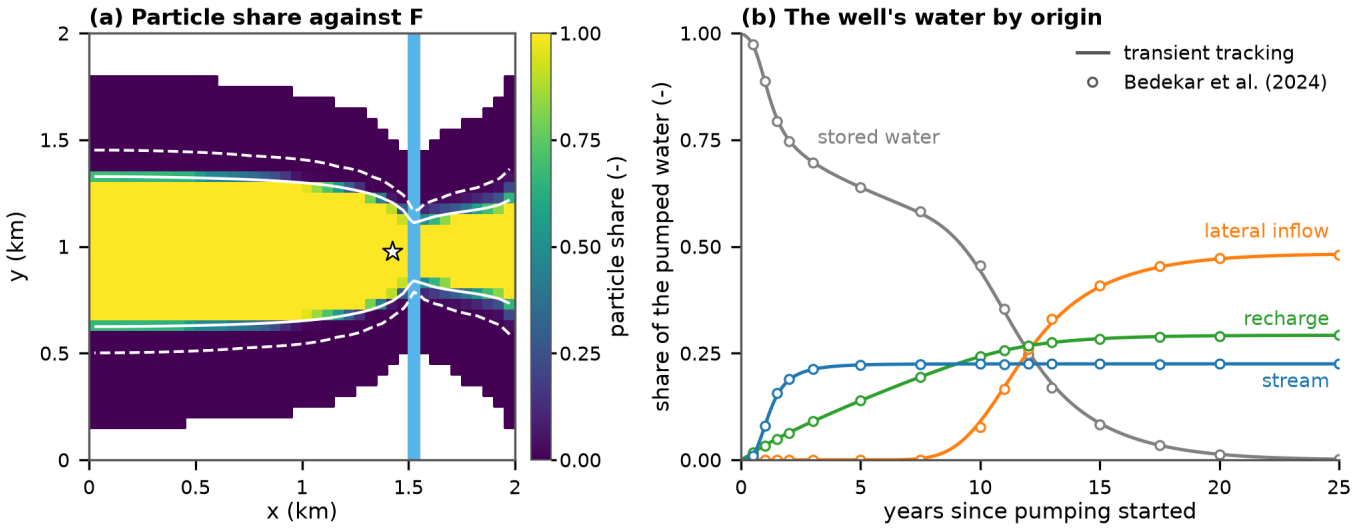


FIGURE 3 Comparisons with other methods. (a) Contributing area of worked example A against weight-losing PRT particles on the same flows. The colours show the particle share in each cell of the example model where the particles start. The particle share is the share of the water released in a cell that the particles deliver to the well. Cells without particles are blank. White lines: F = 0.1 (dashed) and F = 0.5 (solid) from full mixing. Star: well. Blue cells: river. (b) The well's water by origin in the unit-concentration study of Bedekar et al. (2024), rebuilt in MODFLOW 6. Stored water is the water in the aquifer at the start of the run, one day before pumping starts. It is the initial water of Section 2. Lines: forward transient tracking. Open circles: values read off Figure 2 of Bedekar et al. (2024).

TABLE 1 The eleven notebooks that ship with the code and how each compares with another approach (Supporting Information, Text S2).

Notebook	Question	Output	Compared with (agreement)	Fig. 2 panel
01 (A, C)	Where does a well's water come from?	F map, F > 0.5 and F > 0.1 areas, the well's water by source	Weight-losing PRT particles (F > 0.5 area 2.1 % smaller)	a, b
02	What is computed, and what changes at a weak sink?	Gross and net accounting at a weak sink, fractions to every sink	Foley and Black (2013), reproduced to 1×10^{-14}	–
03 (A, C)	Where does a source's water go?	G maps, fate of the recharge on a field	GWT with tagged water, run to equilibrium (to 4.2×10^{-13})	c
04	Does it work on my grid?	F maps and areas on regular (DIS), Voronoi (DISV) and fully unstructured (DISU) grids	The same model on the regular grid (DISU identical, DISV expected area +0.4 %)	h
05 (B)	How does the area change through the seasons?	Seasonal snapshots of the contributing area over a 3-year run	–	e
06 (C)	How old is the water reaching the well?	Mean travel time map, 5- and 25-year zones, mean travel time by source	Weight-losing PRT particles (overall mean travel time within 0.5 %)	d
07 (C)	What if water passes through streams, drains, wells or linked models?	The well's water by source, with the stream reaches of the Streamflow Routing (SFR) package traced as cells of their own or not. A multi-aquifer well. A model split into three linked models	The same model unsplit (to 2.1×10^{-13})	–
08 (B)	How much of the pumped water was already in the aquifer?	The well's water by origin through time	GWT with tagged recharge (to 9.8×10^{-14})	f
09	How does parameter uncertainty change the area?	Probability of contribution over 50 parameter sets, before and after history matching with PEST++ IES (White 2018, White et al. 2020)	A synthetic true model (93 % of the true F > 0.5 area covered after history matching, 79 % before)	g
10 (B)	Does it reproduce a published unit-concentration study?	The well's water by source over 25 years. Time-limited contributing areas from backward tracking	Bedekar et al. (2024), Figure 2 (within 1.3 percentage points), and four GWT runs (to 1.5×10^{-10})	–
11	How wide is the edge of an area, and how does it change with the grid?	Width of the band where F lies between 0.1 and 0.9 against cell size and grid orientation, on two test problems	The analytical capture zone of Javandel and Tsang (1986) and PRT on a fine grid (F > 0.5 area within 4.5 % of the reference area for cells of 50 m or less)	–

On an Apple M4 Max with 64 GB of memory, one trace of nine destination groups (each well alone and all eight together) took 2–3 s and peaked at 2.5 GB of memory. Each further group added about 0.19 GB (Supporting Information, Text S3). These times do not compare like with like, because a `gwsource` trace computes F in every cell while PRT computes particle shares only where particles are released.

The full-mixing areas differ from the published area. We measure this by how much of the published area each delineated area covers. Coverage alone can mislead, because a larger area covers more of the published area. We therefore also compare the size of each area with the size of the published area. The published area comes from one reasonable set of particle settings for where recharge and stream flows enter a cell, where particles stop and how particles that start in dry cells are handled. PRT with the same settings reproduces the published area. Its area covers all 980 cells of the published area and only 2 more cells. Other reasonable settings for particle runs that conserve water already change the area considerably. With the MODFLOW 6 default settings for these flows and for dry start cells, PRT from the same start cells covers only 67 % of the published area. Its area is also a quarter smaller (Supporting Information, Text S3).

Full mixing matches the published area less well than any of these particle runs. The area where F exceeds 0.5 covers 43 % of the published area and is less than half as large. We infer that the main reason is the one seen in the comparison with particles in Section 3. Both methods use the same flows across cell faces. Inside a cell, a particle follows a velocity interpolated from these flows, so where it enters the cell decides where it leaves. Full mixing does not keep this position. It splits the cell's water among all its outflows in proportion to their size. On average, the cells of the published area send 42 % of their water to the wells. Many of these cells therefore fall below F = 0.5 (Fig. 4a). The F > 0.1 area covers 73 % of the published area, but it is about twice its size. Fig. 4b shows this for a window around four of the wells, where the F > 0.1 area covers 538 of the 688 cells of the published area.

Weak sinks add to the difference. A buried stream cell holds a stream reach and has another active cell above it. Where such a reach gains water, it is a weak sink that stops particles. In the published runs, 26,828 particles stopped in buried stream cells and 466 in the five priority wells that take only part of their cell's outflow. Each stopped particle counts in full as reaching that sink, while full mixing passes the rest of the cell's water on (Supporting Information, Text S3). Net accounting attributes to a weak-sink well all the water that enters its cell. With net accounting, the F > 0.5 area covers 46 % of the published area, against 43 % with gross accounting. At the weak-sink wells, gross accounting comes closer to the size of each well's published area than net accounting does.

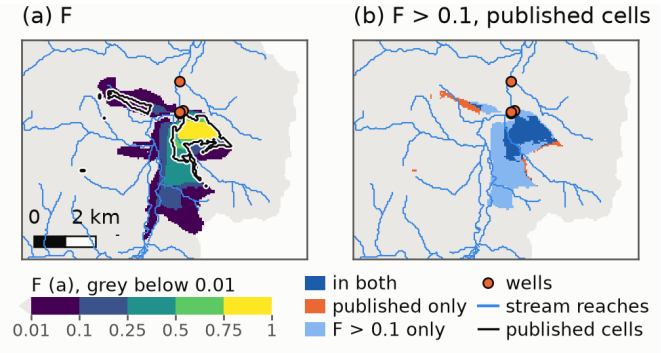


FIGURE 4 F and the published contributing area of the Neversink model, in a 7.1 × 9.25 km window around four of the priority wells. The window holds 688 of the 980 cells of the published area. (a) F in the calibrated model, with the published area outlined. (b) The F > 0.1 area against the published area. Dark blue: in both (538 cells). Orange: published area only (150). Light blue: F > 0.1 only (1,090).

Example C (Section 3) showed that a sink's split by source depends on how stream reaches are traced. On this model the split also depends on the method. With the stream reaches left as source and sink terms, full mixing attributes 57 % of the wells' water to recharge and 43 % to stream leakage. With the reaches traced as cells of their own, all of the wells' water traces back to recharge. Particle runs that place particles from the full-mixing trace and lose weight at sinks as full mixing does split the wells' water 49 % recharge and 51 % stream leakage, with the reaches again left as source and sink terms. Both methods deliver the wells' total inflow to within 2 %. We infer, without tracing it cell by cell, that stream water leaking near the wells stays on particle paths that end at a well. Full mixing instead assigns that water the average fate of all the water leaving the cells it passes. A split by source on a regional model should therefore be reported with the method that produced it.

5. TRADE-OFFS AND CONCLUSIONS

From the flows that a MODFLOW 6 model saves, `gwsource` maps where the water of any sink comes from and where the water of any source goes. The same trace splits a sink's water by source and computes mean travel times. The code extends the full-mixing method of FlowSource (Foley and Black 2013) to every MODFLOW 6 grid type, to stream networks, to linked models and to transient flow. Every result is checked against the water budget (Section 2). The outputs agree with GWT and with a published unit-concentration study (Section 3). On a published regional model, a trace took less time than the fastest particle-tracking run (Section 4).

The main trade-off is that full-mixing areas are wider and more diffuse than particle areas (Section 2). Their edge is a band of cells where F lies between 0 and 1. Finer cells and a grid aligned with the main flow narrow this band (Supporting Information, Text S4). No single threshold on F therefore reproduces a particle area everywhere. A particle area is not a fixed reference either, because it depends on its particle settings as much as on the flow model (Section 4). The chosen threshold is a judgement about what the area is for. For this reason, the chosen threshold should be reported with the area, along with the accounting and how stream reaches were traced (Section 3).

A sink's overall mean travel time and its split by source are less sensitive to mixing than the edge of its area is. Any method that conserves water yields the same overall mean travel time (Section 3). In a test problem, particles produced the same split by source (Supporting Information, Text S4). On the regional model, with its more complex flow system, the split by source differed between the two methods (Section 4).

We would therefore use particles where a sharp edge or a first arrival matters, such as a short time-of-travel zone. We would use GWT where dispersion, sorption or decay matter, because transient tracking moves water by advection only (Langevin et al. 2022). Full mixing suits many sinks, unstructured grids, package networks and repeated runs, such as one trace for every realisation of an ensemble.

The method also has practical limits. Transient tracking has been tested on synthetic models only. Its memory grows with the number of time steps. Movers between models (the MVE package) are not implemented yet. Without cell vertices, the code cannot map results on a DISU grid.

`gwsource` is an open-source Python package (MIT licence) that reads the binary grid and budget files of a MODFLOW 6 model on any grid type. It is tested on ten small test models and two published examples (Section 6).

6. CODE AND DATA AVAILABILITY

The code, `gwsource`, is open source under the MIT licence at https://github.com/rhugman/gw-source [check: public URL]. This paper uses version 0.0.1 [check: release version and tag]. The eleven notebooks of Table 1 ship with the release under the same numbers. The same release holds the verification tables (docs/verification.md), the test models (tests/models/README.md) and the notes of the Neversink study (docs/neversink_study.md). The Neversink model is in a USGS data release (Fienen and Corson-Dosch 2021). The files behind every figure will be archived on Zenodo [check: DOI].

REFERENCES

Bedekar, V., Neville, C., Tonkin, M.J., Bartlett, R.D. & Plato, P. (2024) A unit-concentration method to quantify source contribution. *Groundwater*, 62(2), 303–309. doi:10.1111/gwat.13333.

Cornaton, F. & Perrochet, P. (2006) Groundwater age, life expectancy and transit time distributions in advective–dispersive systems: 1. Generalized reservoir theory. *Advances in Water Resources*, 29(9), 1267–1291. doi:10.1016/j.advwatres.2005.10.009.

Corson-Dosch, N.T., Fienen, M.N., Finkelstein, J.S., Leaf, A.T., White, J.T., Woda, J. et al. (2022) *Areas contributing recharge to priority wells in valley-fill aquifers in the Neversink River and Rondout Creek drainage basins, New York*. U.S. Geological Survey, Reston, Virginia. Scientific Investigations Report 2021–5112.
URL https://doi.org/10.3133/sir20215112

Environment Agency (2019) *Manual for the Production of Groundwater Source Protection Zones*. Bristol, UK: Environment Agency.
URL https://assets.publishing.service.gov.uk/media/5d41a020e5274a0a0bf7757c/Manual-for-the-production-of-Groundwater-Source-Protection-Zones.pdf

Fienen, M.N. & Corson-Dosch, N.T. (2021) *Groundwater model archive and workflow for Neversink/Rondout Basin, New York, source water delineation*. U.S. Geological Survey data release.
URL https://doi.org/10.5066/P9HWSOHP

Fienen, M.N., Corson-Dosch, N.T., White, J.T., Leaf, A.T. & Hunt, R.J. (2022) Risk-based wellhead protection decision support: A repeatable workflow approach. *Groundwater*, 60(1), 71–86. doi:10.1111/gwat.13129.

Foley, C. & Black, A. Efficiently delineating volumetric capture areas and flow pathways using directed acyclic graphs and MODFLOW; description of the algorithms within FlowSource. In: *Proceedings of MODFLOW and More 2013: Translating Science into Practice, Golden, Colorado, June 2–5, 2013*. Golden, Colorado: IGWMC, Colorado School of Mines, pp. 571–581, [check: pages against the proceedings].
URL https://groundwater-science.co.uk/wp-content/uploads/2013_FSv6_DAGPaper.pdf

Goode, D.J. (1996) Direct simulation of groundwater age. *Water Resources Research*, 32(2), 289–296. doi:10.1029/95WR03401.

Javandel, I. & Tsang, C.-F. (1986) Capture-zone type curves: A tool for aquifer cleanup. *Ground Water*, 24(5), 616–625. doi:10.1111/j.1745-6584.1986.tb03710.x.

Langevin, C.D., Hughes, J.D., Banta, E.R., Niswonger, R.G., Panday, S. & Provost, A.M. (2017) *Documentation for the MODFLOW 6 Groundwater Flow Model*. U.S. Geological Survey, Reston, Virginia. Techniques and Methods book 6, chap. A55.
URL https://doi.org/10.3133/tm6A55

Langevin, C.D., Hughes, J.D., Provost, A.M., Russcher, M.J., Reno, M.J., Bonelli, W.P. et al. (2026) *MODFLOW 6 Modular Hydrologic Model version 6.8.1*. U.S. Geological Survey Software Release, 23 September 2026.
URL https://doi.org/10.5066/P1PGE9XW

Langevin, C.D., Provost, A.M., Panday, S. & Hughes, J.D. (2022) *Documentation for the MODFLOW 6 Groundwater Transport Model*. U.S. Geological Survey, Reston, Virginia. Techniques and Methods book 6, chap. A61.
URL https://doi.org/10.3133/tm6A61

Neupauer, R.M. & Wilson, J.L. (1999) Adjoint method for obtaining backward-in-time location and travel time probabilities of a conservative groundwater contaminant. *Water Resources Research*, 35(11), 3389–3398. doi:10.1029/1999WR900190.

Partington, D., Brunner, P., Simmons, C.T., Therrien, R., Werner, A.D., Dandy, G.C. et al. (2011) A hydraulic mixing-cell method to quantify the groundwater component of streamflow within spatially distributed fully integrated surface water–groundwater flow models. *Environmental Modelling & Software*, 26(7), 886–898. doi:10.1016/j.envsoft.2011.02.007.

Pollock, D.W. (2016) *User guide for MODPATH Version 7—A particle-tracking model for MODFLOW*. U.S. Geological Survey, Reston,

Virginia. Open-File Report 2016–1086.
URL https://doi.org/10.3133/ofr20161086

Potter, S.T., Moreno-Barbero, E. & Divine, C.E. (2008) MODALL: A practical tool for designing and optimizing capture systems. *Ground Water*, 46(2), 335–340. doi:10.1111/j.1745-6584.2007.00410.x.

Provost, A.M., Langevin, C.D. & Hughes, J.D. (2017) *Documentation for the "XT3D" option in the Node Property Flow (NPF) Package of MODFLOW 6*. U.S. Geological Survey, Reston, Virginia. Techniques and Methods book 6, chap. A56.
URL https://doi.org/10.3133/tm6A56

White, J.T. (2018) A model-independent iterative ensemble smoother for efficient history-matching and uncertainty quantification in very high dimensions. *Environmental Modelling & Software*, 109, 191–201. doi:10.1016/j.envsoft.2018.06.009.

White, J.T., Hunt, R.J., Fienen, M.N. & Doherty, J.E. (2020) *Approaches to highly parameterized inversion: PEST++ Version 5, a software suite for parameter estimation, uncertainty analysis, management optimization and sensitivity analysis*. U.S. Geological Survey, Reston, Virginia. Techniques and Methods 7C26.
URL https://doi.org/10.3133/tm7C26